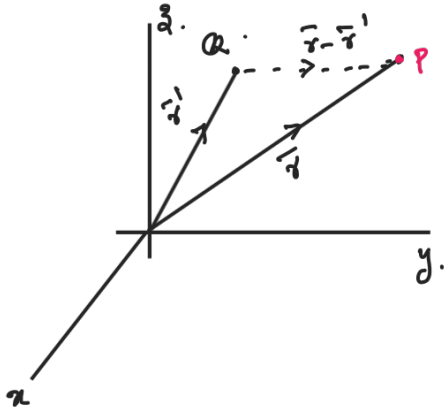


Mathematical tool to find electric field from electrostatic potential

$$\phi(\vec{r}) = \frac{Q}{4\pi\epsilon_0} \frac{1}{|\vec{r} - \vec{r}'|}$$

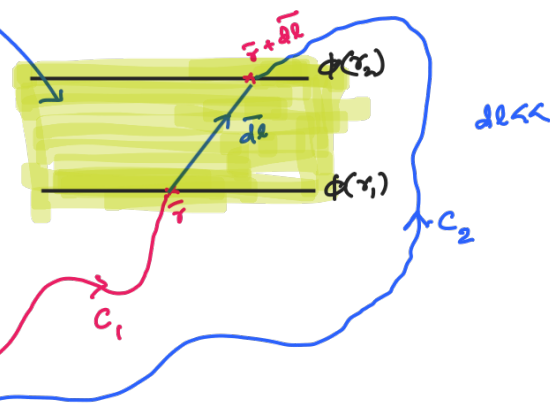
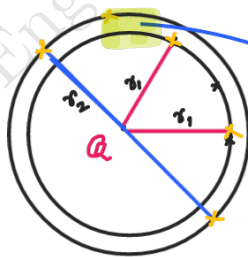
$\phi(\vec{r}) \rightarrow$ ① Scalar field $\Rightarrow$ It is a number associated with every position $\vec{r}$.

② It is derived from an integral $\Rightarrow$ it should be a smooth function $\Rightarrow$ It should be continuous and differentiable.

Example: For a single point charge Q at origin

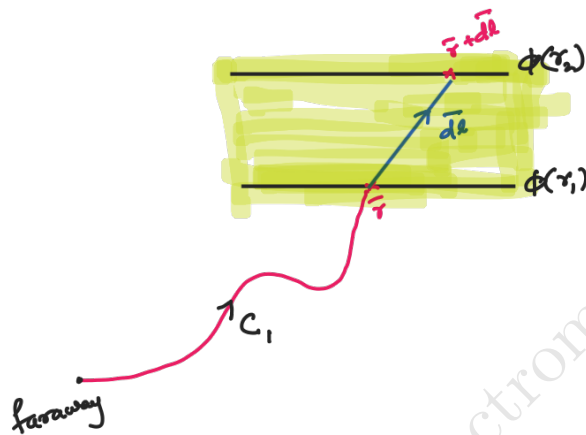
$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

Equipotential level curves are concentric circles.



$$\phi(\vec{r}) = - \int_{C_1} \vec{E}(\vec{r}) \cdot d\vec{r} \quad \dots (1)$$

$$\begin{aligned} \phi(\vec{r} + d\vec{r}) &= - \int_{C_2} \vec{E}(\vec{r}) \cdot d\vec{r} = - \int_{C_1} \vec{E}(\vec{r}) \cdot d\vec{r} + (- \vec{E}(\vec{r}) \cdot d\vec{r}) \\ &= \phi(\vec{r}) - \vec{E}(\vec{r}) \cdot d\vec{r} \end{aligned}$$



$$\Rightarrow \phi(\vec{r} + d\vec{r}) - \phi(\vec{r}) = - \vec{E}(\vec{r}) \cdot d\vec{r}$$

$$\phi(r_2) - \phi(r_1) = - \vec{E}(\vec{r}) \cdot d\vec{r}$$

$$\Delta\phi = - \vec{E}(\vec{r}) \cdot d\vec{r}$$

$$\frac{\Delta\phi}{d\vec{r}} = - \vec{E}(\vec{r}) \cdot \frac{d\vec{r}}{d\vec{r}}$$

$$\approx \frac{d\phi}{d\vec{r}} = - \vec{E}(\vec{r}) \cdot \frac{d\vec{r}}{d\vec{r}} \quad \dots (2)$$

(Ref Lec 9
EE1203
Vector
Calculus)

→ But by the definition of
directional derivative

$$\frac{d\phi}{d\vec{r}} = \nabla\phi \cdot \frac{d\vec{r}}{d\vec{r}} \quad \dots (3)$$

$$\Rightarrow \boxed{\vec{E} = - \nabla\phi}$$

$\nabla\phi \rightarrow$ Gradient of
scalar potential.

* Example problem: Find out the electric field due to a
dipole $(+q \text{ --- } -q)$ using the gradient of
electrostatic potential.

Concept of Electric flux & Gauss's law (Maxwell's 1st Equation)

* Revise Lect-15 & 16 of EE1203 (Vector Calculus)
(Concept of flux of a vector field & divergence theorem)

Electric flux: Total field lines passing through

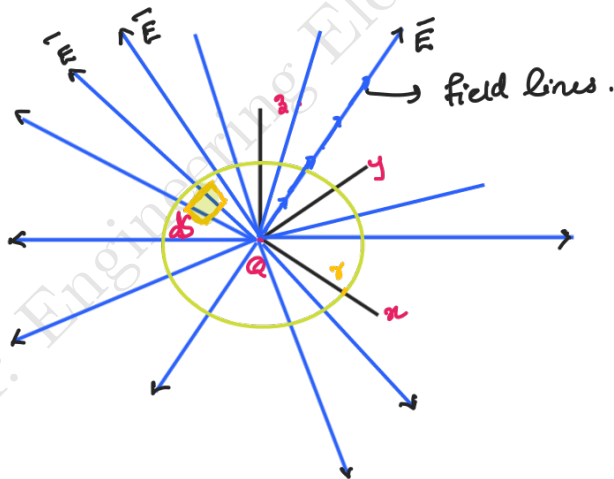
$$\text{Electric flux} = \int_S \vec{E} \cdot \hat{n} dS$$

For a point charge ;

$$\vec{E} = \frac{Q}{4\pi\epsilon_0} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

Case 1

Let us assume a point charge at the origin.



If we want to calculate how much electric field leaves this sphere of radius 'r'?

$$\begin{aligned}
 \iint \vec{E} \cdot \hat{n} \, dS &= \iint E_r \, dS \quad \left(\because \begin{array}{l} \vec{E} = E_r \hat{r} \\ \hat{n} = \hat{r} \end{array} \right) \\
 &= \frac{Q}{4\pi\epsilon_0} \iint \frac{1}{r^2} \, dS \\
 &= \frac{Q}{4\pi\epsilon_0} \iint \frac{1}{r^2} \cdot r^2 \sin\theta \, d\theta \, d\phi \\
 &= \frac{Q}{4\pi\epsilon_0} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin\theta \, d\theta \, d\phi = \frac{Q}{\epsilon_0}
 \end{aligned}$$

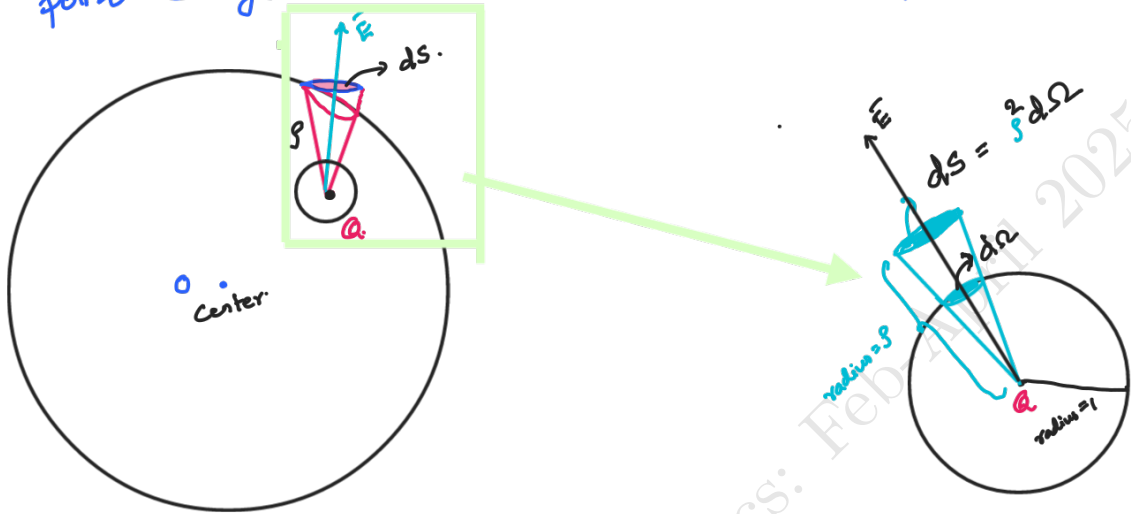
$dS_r \rightarrow r^2 \sin\theta \, d\theta \, d\phi$
 $\theta \rightarrow$ angle extended w.r.t z-axis.
 $\phi \rightarrow$ Azimuthal angle
(independent of radius 'r')

$\Rightarrow$ Regardless of the radius 'r' of the sphere we consider, the total electric field leaving the closed surface depends only on how much charge is inside the volume enclosed $\left(= \frac{Q}{\epsilon_0} \right)$.

Electric field is $\propto$ number of field lines/unit area.

$\Rightarrow \int_S \vec{E} \cdot \hat{n} \, dS \Rightarrow$ total number of field lines leaving the surface 'S'.

Case I
* If the point charge is not at the center of the sphere?



$$\oint_S \vec{E} \cdot \hat{n} dS = ?$$

$$\vec{E} \cdot \hat{n} dS = \frac{Q}{4\pi\epsilon_0} \times \frac{1}{s^2} \underbrace{s^2 d\Omega}_{=dS}$$

$$= \frac{Q}{4\pi\epsilon_0} d\Omega$$

$$\oint \vec{E} \cdot \hat{n} dS = \frac{Q}{4\pi\epsilon_0} \iint d\Omega = \frac{Q}{4\pi\epsilon_0} \int_0^\pi \int_0^{2\pi} \underbrace{\sin\theta d\theta d\phi}_{d\Omega}$$

$$= \frac{Q}{4\pi\epsilon_0} \times 4\pi = \frac{Q}{\epsilon_0}$$

$$d\Omega = \frac{1}{r^2} \sin\theta d\theta d\phi$$

$$r=1$$

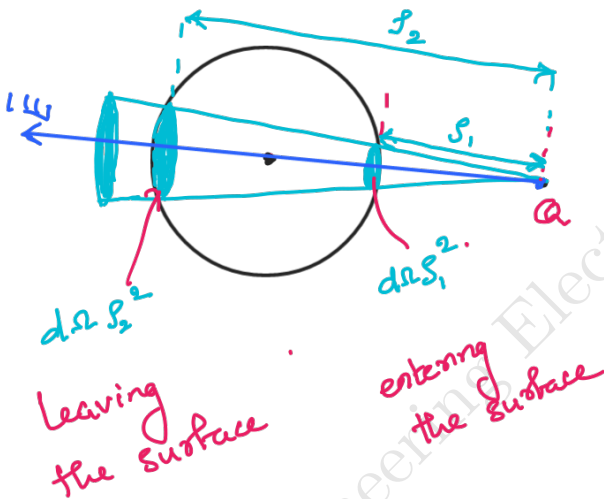
This idea says $\Rightarrow$ whenever we put the point charge in the closed surface the total electric field lines (flux) remains the same and it depends only on the total charge Q .

In the above example, we worked with solid angle (not constrained to sphere only). So the above conclusion can be generalized to any closed surface.

$$\Rightarrow \oint_S \vec{E} \cdot \hat{n} dS = \frac{Q}{\epsilon_0}.$$

Case III

Charge outside the sphere.



$$\text{Flux entering} = \frac{Q}{4\pi\epsilon_0} \frac{1}{r_1^2} \times d\Omega S_1^2$$

$$\text{Flux leaving} = \frac{Q}{4\pi\epsilon_0} \frac{1}{r_2^2} \times d\Omega S_2^2$$

$$\Rightarrow \text{Total Flux} = \oint_S \vec{E} \cdot \hat{n} dS = 0.$$

In summary:

$$\oint_S \vec{E} \cdot \hat{n} dS = \begin{cases} 0 & \text{if } Q \text{ not in volume.} \\ \frac{Q}{\epsilon_0} & \text{if } Q \text{ in volume.} \end{cases}$$

$\Rightarrow$ Maxwell's 1st Equation in integral form.

Now By divergence theorem;

$$\oint \vec{F} \cdot \hat{n} ds = \int_{vol} (\nabla \cdot \vec{F}) dv$$

$$\Rightarrow \oint \vec{E} \cdot \hat{n} ds = \int_{vol} (\nabla \cdot \vec{E}) dv = \frac{Q}{\epsilon_0}$$

But in real life we know we have cloud of charges over a region rather than point charge.

$$Q = \underbrace{\frac{Q}{\Delta V}}_P \cdot \Delta V = \rho \cdot dv \quad (\Delta V \rightarrow \infty)$$

charge density

$$\Rightarrow \int_{vol} (\nabla \cdot \vec{E}) dv = \frac{1}{\epsilon_0} \int_{vol} \rho dv$$

$$\Rightarrow \boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}} \leftarrow \text{differential form of Maxwell's first equation.}$$

